



tomers. We don't do upfront sales, but we come with the entire platform of energy services, vehicle servicing, and maintenance," he says.

Although EV concepts are coming up in a big way, Naskar feels there is a huge gap in understanding and customers are mostly hesitant in adopting EV technologies. He says, "We thought to match that gap by holistic approaches. If someone wants to shift towards electric vehicles, they should not be worrying about other things like maintenance of the vehicle or energy infrastructures."

According to India Energy Storage Alliance (IESA), the Indian EV market is estimated to grow at a CAGR of 36% annually till 2026. The Indian automotive industry is the fifth largest in the world and is slated to be the third largest by 2030. By making the shift towards EVs, India stands to benefit on many fronts due to its relative abundance of renewable energy resources and availability of skilled manpower in the technology and manufacturing sectors.

From a traditional electrical system manufacturing group, Chedda Electronics, Abhay Patwardhan, Director, EV Business, says his company decided to work in the back end of the technology rather than going into vehicle business. He adds, "So, we started working on the battery management system (BMS), the battery packs, swapping solutions, and almost all back ends."

Though not directly into battery technology, Sangeet Kumar, CEO, Addverb Technologies, is working on robotics that uses different types of batteries. "Currently we are buying from a US company, but every US company is manufacturing in China. We are now working with one of the largest players in India. We hope it would work as we have now understood a lot about BMS, the electronics, and the software pack," he says.

Sharing it as a 'pain point,' Kumar says manufacturing batteries had always been with Chinese compa-

nies, be it in the European market, or in the US market. "The battery is one of the key components used in all kinds of products. We in India do not have that kind of support system or manufacturing system that is there in China, though we are far better than them as far as software is concerned," Kumar opines.

Battery tech in India

There's no shortage of excitement for EV battery startups or investments in the industry, as companies and design engineers are looking to reap the opportunities. The Central government has announced a production-linked incentive (PLI) of 180 billion rupees for developing battery technologies in India. We are in a position to develop a complete ecosystem that is required for the high demand of EVs in India.

Naskar feels the PLI schemes will definitely push the local manufacturing sector to work in building technology solutions. "A lot of MS-MEs will participate in the market and start building products for very specific applications," he says.

However, the PLI scheme is not really designed to support a complete startup ecosystem, because it needs some kind of capital investment first. "But anyhow, it is definitely an encouraging scheme. And once big companies come into this game in a big way, there will be a lot of supporting ecosystem, and the market will be open for a lot more innovation. It is a good step towards improving the overall technology landscape that lies for lithium-ion batteries in the country," Naskar explains.

Gupta says, "In order of magnitude, we are probably ahead compared to three years ago. So, the industry has actually taken significant steps. The maturity seen in battery pack assemblers is now also seen in people raising questions about it. So that has improved tremendously. But we continue to fall behind some

of the other countries. And that essentially means, there is a significant gap for us to improve."

Beginning with Tier I companies, Gupta says, the process of maturity in battery assembly is just a fraction of what you see in some of the other product categories. He opines, "A couple of reasons for that may be, for you to achieve process maturity, it requires significant volume and scale. The entire Indian industry doesn't have a volume and scale where you could actually go for a proper grading of the cells, which are imported." He adds, "If I aggregate the entire volume the Indian industry has, it will probably be smaller than the volume of a single large plant in China."

"Despite the overwhelming issues, the positive side is we are significantly ahead of what we were in 2017 and 2018. So, there is a significant amount of hope for the industry. But it is also important to note, we are still behind some of the other countries. Even Tier 3 manufacturers in China probably are better than Tier 1 manufacturers in India," Gupta opines.

From the usage perspective, Patwardhan says, either technology or the batteries will have to be sourced, and in application perspective, we are putting the costs in front of the technology. He says, "The intention of the incentive is basically to promote the internal R&D development, so that the cost is pushed behind and technology comes into play. But still, that seems to be a bit of a concern today."

Patwardhan says, for attaining the quality today, we have to understand the best assembly of the cells. According to him, "We should make the best packs out of the cells and then go for cell manufacturing as there are well established resources and processes. And then the entire ecosystem will be able to get the right kind of product suitable for the Indian market."